Critical appraisal

TAK-003 tetravalent dengue vaccine

4.5-year phase 3 randomized, double-blind, placebo-controlled trial

20,099	**8**	**4.5 y**	**2:1**
randomized participants	endemic countries	follow-up after dose 2	TAK-003 : placebo

Study: Tricou, Yu, Reynales et al. Lancet Global Health 2024

Critical appraisal verdict first

Outcome-specific appraisal using CONSORT and RoB 2 logic

Overall interpretation

A strong trial for the broad outcomes of symptomatic virologically confirmed dengue and hospitalised dengue in children/adolescents in endemic settings.

But not a clean all-serotype story

Evidence is weaker for dengue-naive participants against DENV-3 and DENV-4. Subgroup and severe-disease analyses have sparse events and wide intervals.

Risk of bias: low for overall VCD

Randomization, allocation concealment, blinding, centralized RT-PCR outcome confirmation, and high follow-up completion support internal validity.

Some concerns for hospitalisation/severe endpoints

Hospitalization thresholds differed by country; Sri Lanka admitted a much higher proportion of dengue cases. This affects clinical severity interpretation.

Subgroups: hypothesis-generating

Serostatus and serotype findings are clinically important, but many comparisons were exploratory and underpowered after stratification.

Bottom line: useful vaccine signal, with DENV-3/DENV-4 seronegative uncertainty

Tricou et al. Lancet Global Health 2024; supplementary appendix 4; CONSORT/RoB 2 appraisal lens.

PICO and clinical question

What problem did the trial try to solve?

PICO element	In this trial	Appraisal comment
Population	Healthy children/adolescents aged 4-16 years in endemic countries	Relevant to pediatric public health; less direct for adults/travellers
Intervention	Two subcutaneous TAK-003 doses, 3 months apart	Clear, reproducible schedule
Comparator	Saline placebo	Appropriate for efficacy and safety
Outcomes	RT-PCR-confirmed symptomatic dengue; hospitalised VCD; serious adverse events	Hard virological outcome; clinical severity outcome more context-dependent

Question answered well

Does TAK-003 reduce virologically confirmed dengue and hospitalised dengue over several years in endemic pediatric populations?

Question not fully answered

Does it protect dengue-naive recipients equally against all 4 serotypes, especially DENV-3 and DENV-4?

Tricou et al. Lancet Global Health 2024; supplementary appendix 4; CONSORT/RoB 2 appraisal lens.

Why this matters

Dengue vaccine development is unusually hard

4
DENV serotypes requiring balanced protection

2
key safety contexts: seropositive vs seronegative

years
needed to detect waning and late harm

policy
requires endemic-setting benefit-risk

Clinical background

Previous dengue exposure changes the risk-benefit calculation.
CYD-TDV experience made seronegative safety a central question.
Programmes need durable protection, not only short-term cross-protection.
Hospitalisation prevention can matter more than mild case prevention.

Appraisal focus

Does randomisation protect the comparison?
Are outcomes clinically meaningful and measured consistently?
Are subgroup claims credible?
Can these data be applied to local vaccination policy?

Current WHO recommendations were checked for context; see references.

Study design at a glance

Multicountry randomized efficacy trial

Design
Phase 3, randomized, double-blind, placebo-controlled, parallel group trial

Sites
26 medical/research centres across Brazil, Colombia, Dominican Republic, Nicaragua, Panama, Philippines, Sri Lanka, Thailand

Population
Healthy participants aged 4-16 years; baseline serostatus measured by microneutralisation

Allocation
2:1 TAK-003 to placebo; stratified by age and region; interactive web response system

Follow-up
Weekly febrile illness contacts with RT-PCR testing; serious adverse events throughout

Analysis
Cox proportional hazards model; exploratory cumulative efficacy up to about month 57

Strong design for vaccine efficacy - but exploratory subgroup interpretation needs restraint

Tricou et al. Lancet Global Health 2024; supplementary appendix 4; CONSORT/RoB 2 appraisal lens.

Participant selection and baseline balance

Who was studied?

	Safety set		Per protocol set	
	Placebo (n=6687)	TAK-003 (n=13 380)	Placebo (n=6317)	TAK-003 (n=12 704)
Age, mean (SD), years	9·6 (3·34)	9·6 (3·36)	9·6 (3·34)	9·6 (3·35)
Age group				
4–5 years	846 (12·7%)	1702 (12·7%)	801 (12·7%)	1620 (12·8%)
6–11 years	3697 (55·3%)	7387 (55·2%)	3492 (55·3%)	7010 (55·2%)
12–16 years	2144 (32·1%)	4291 (32·1%)	2024 (32·0%)	4074 (32·1%)
Sex				
Male	3411 (51·0%)	6729 (50·3%)	3219 (51·0%)	6390 (50·3%)
Female	3276 (49·0%)	6651 (49·7%)	3098 (49·0%)	6314 (49·7%)
Race*				
American Indian or Alaska Native	2621 (39·2%)	5234 (39·1%)	2378 (37·6%)	4819 (37·9%)
Asian	2985 (44·6%)	5988 (44·8%)	2934 (46·4%)	5888 (46·3%)
Black or African American	758 (11·3%)	1451 (10·8%)	706 (11·2%)	1351 (10·6%)
Native Hawaiian or other Pacific Islander	1 (<0·1%)	2 (<0·1%)	1 (<0·1%)	2 (<0·1%)
White	149 (2·2%)	329 (2·5%)	131 (2·1%)	284 (2·2%)
Multiracial	170 (2·5%)	376 (2·8%)	165 (2·6%)	360 (2·8%)
Region				
Asia	2993 (44·8%)	5996 (44·8%)	2942 (46·6%)	5896 (46·4%)
Latin America	3694 (55·2%)	7384 (55·2%)	3375 (53·4%)	6808 (53·6%)
Baseline serostatus, n/N (%)				
Seropositive	4854/6684 (72·6%)	9663/13 375 (72·2%)	4588/6314 (72·7%)	9167/12 700 (72·2%)
Seronegative	1832/6684 (27·4%)	3714/13 375 (27·8%)	1726/6314 (27·3%)	3533/12 700 (27·8%)

Data are n (%) unless otherwise stated. *Data missing for three participants (<0·1%) in the placebo group (safety set data) and two participants (<0·1%) in the placebo group (per protocol set data). For baseline serostatus, percentages are based on the number of participants/number of evaluable participants. Baseline seropositivity is defined as a reciprocal neutralising titre of 10 or more for at least one dengue virus serotype.

***Table 1:* Baseline characteristics**

Table 1 crop: baseline characteristics

Key points

Mean age: 9.6 years.
Sex balance roughly equal.
Asia 44.8%; Latin America 55.2%.
Baseline seronegative: about 27.6%.
Exclusions: pregnancy, immune impairment, serious chronic disease, prior dengue vaccine.

Trial flow: good retention over long follow-up

CONSORT flow diagram

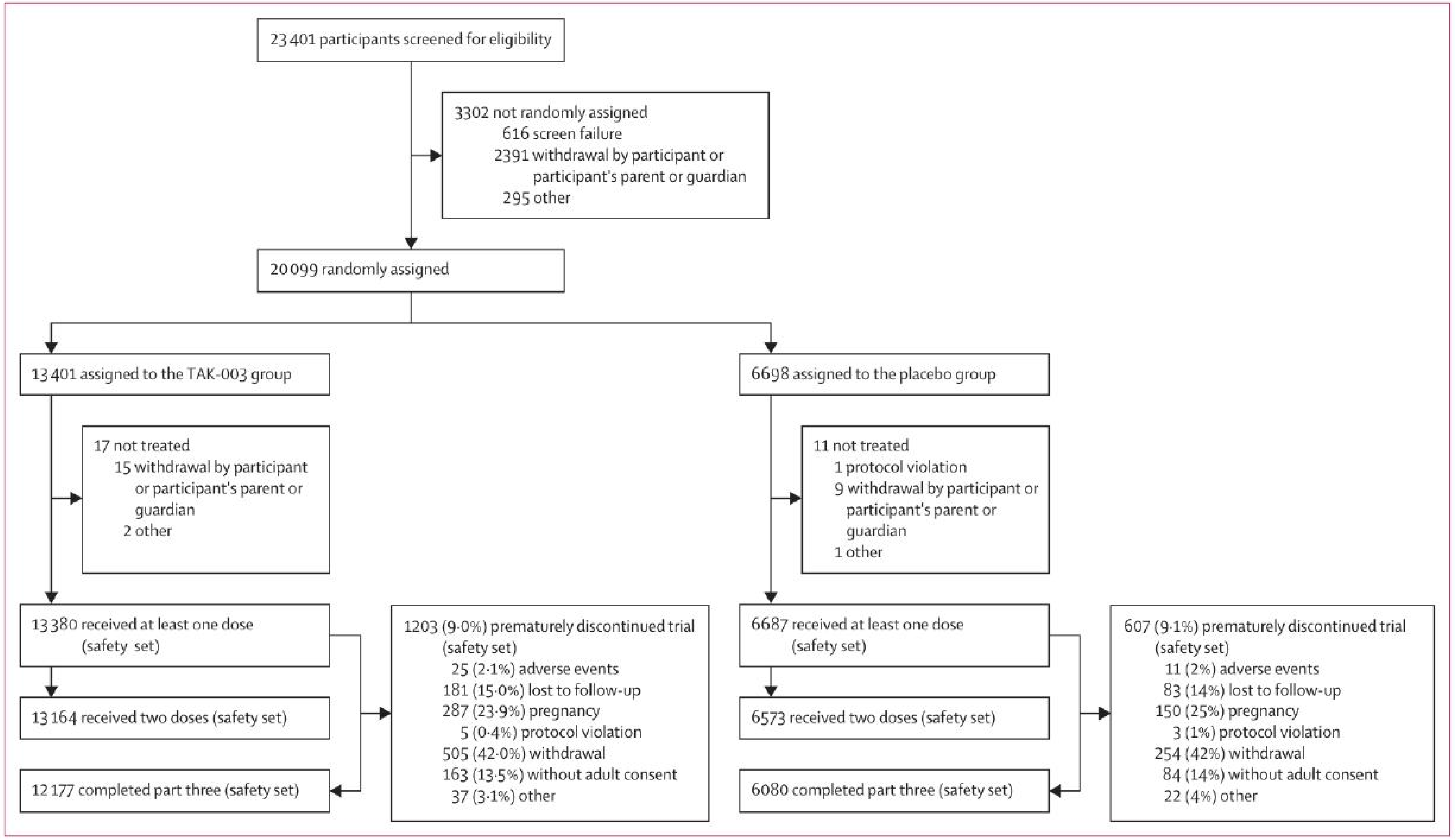


Figure 1: **Trial profile**
Four participants received both TAK-003 and placebo because of an administrative error; these participants are not shown under the TAK-003 or placebo groups.

Appraisal note

Part 3 completion was 91.0%. Discontinuation was similar between arms: 9.0% TAK-003 vs 9.1% placebo.

Intervention and comparator

Biological plausibility and practical delivery

TAK-003
Live-attenuated tetravalent dengue vaccine, based on a DENV-2 backbone. Dose: 0.5 mL subcutaneously at months 0 and 3.

Placebo
0.5 mL saline injection. Blinding preserved for participants, parents/guardians, investigators, and sponsor representatives advising conduct.

Why this is useful

Clear schedule: two doses separated by 3 months.
Placebo comparator allows direct efficacy estimation.
Blinding reduces performance and detection bias.
Subcutaneous administration is feasible for mass vaccination.

What is still unknown

How booster doses modify long-term benefit-risk.
Effectiveness in routine public programmes outside trial centres.
Use in pregnancy, immunocompromise, and age groups outside trial range.

Tricou et al. Lancet Global Health 2024; supplementary appendix 4; CONSORT/RoB 2 appraisal lens.

Outcome ascertainment

A major strength of the trial

Primary clinical signal
Virologically confirmed dengue (VCD): febrile illness plus serotype-specific RT-PCR confirmation.

Safety signal
Serious adverse events and adverse events leading to discontinuation monitored throughout.

27,684
febrile illnesses reported

~98%
febrile episodes tested in full pre-booster period

1,007
VCD cases detected

188
hospitalised VCD cases

Appraisal note

RT-PCR confirmation is objective and serotype-specific.
Weekly surveillance reduces missed febrile episodes.
Hospitalisation is clinically meaningful but depends on local practice.

Tricou et al. Lancet Global Health 2024; supplementary appendix 4; CONSORT/RoB 2 appraisal lens.

Statistical analysis

Sound main framework; caution for exploratory layers

Pre-specified core method

Vaccine efficacy = 1 - hazard ratio.
Cox model adjusted for age and stratified by region.
Primary endpoint powered to rule out VE <=25% at 1 year.
Safety set and per-protocol set both reported.

Limitations of inference

Most 4.5-year subgroup analyses are exploratory.
Multiplicity control was limited.
Serotype-specific and severe disease event counts become small after stratification.
Proportional hazards assumption testing was not prominently reported in the article.

Critical appraisal judgment

Main overall VCD and hospitalised VCD results are credible. Fine-grained serotype/serostatus/severity claims should be treated as less certain.

Overall efficacy: clinically important signal

Cumulative vaccine efficacy from first dose to ~month 57

61.2%

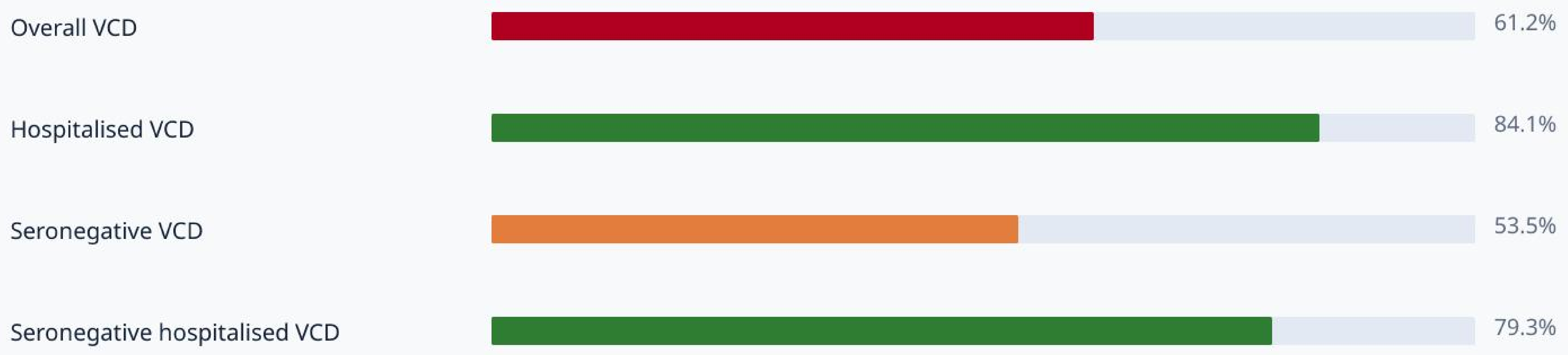
against virologically confirmed dengue (95% CI 56.0-65.8)

84.1%

against hospitalised VCD (95% CI 77.8-88.6)

0

deaths judged vaccine-related



Interpretation

Efficacy against hospitalisation was stronger than efficacy against any symptomatic VCD, suggesting disease modification and clinical impact.

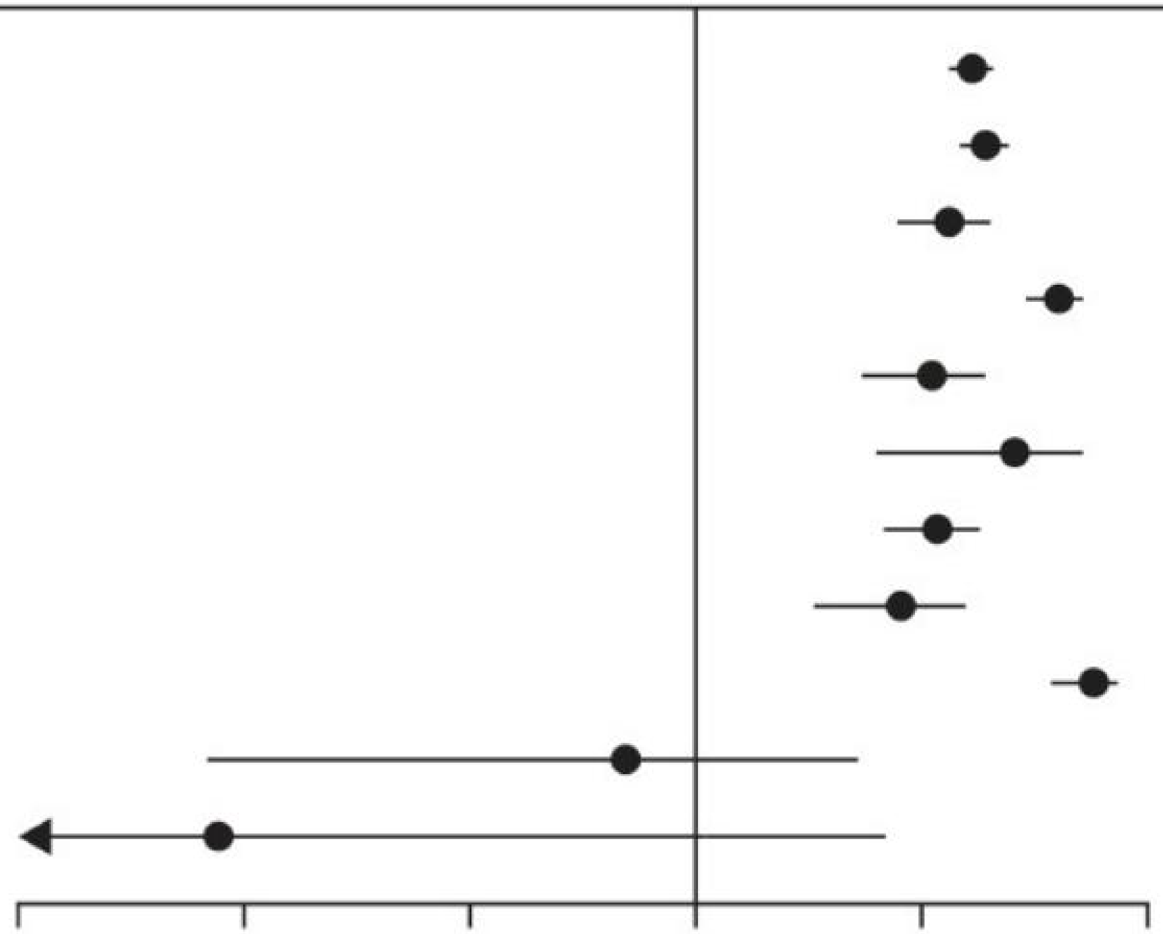
Tricou et al. Lancet Global Health 2024; supplementary appendix 4; CONSORT/RoB 2 appraisal lens.

Forest plot: overall symptomatic dengue

Trial figure crop

A Virologically confirmed dengue

	n/N (%) Placebo	n/N (%) TAK-003	Efficacy, % (95% CI)
Overall	**547/6687 (8·2%)**	**442/13 380 (3·3%)**	**61·2 (56·0 to 65·8)**
Seropositive	394/4854 (8·1%)	295/9663 (3·1%)	64·2 (58·4 to 69·2)
DENV-1	151/4854 (3·1%)	133/9663 (1·4%)	56·1 (44·6 to 65·2)
DENV-2	135/4854 (2·8%)	54/9663 (0·6%)	80·4 (73·1 to 85·7)
DENV-3	97/4854 (2·0%)	96/9663 (1·0%)	52·3 (36·7 to 64·0)
DENV-4	20/4854 (0·4%)	12/9663 (0·1%)	70·6 (39·9 to 85·6)
Seronegative	153/1832 (8·4%)	147/3714 (4·0%)	53·5 (41·6 to 62·9)
DENV-1	79/1832 (4·3%)	89/3714 (2·4%)	45·4 (26·1 to 59·7)
DENV-2	58/1832 (3·2%)	14/3714 (0·4%)	88·1 (78·6 to 93·3)
DENV-3	16/1832 (0·9%)	36/3714 (1·0%)	–15·5 (–108·2 to 35·9)
DENV-4	3/1832 (0·2%)	12/3714 (0·3%)	–105·6 (–628·7 to 42·0)



Read the plot carefully

Clear protection overall and in seropositive participants. In seronegative participants: DENV-1 and DENV-2 protected; DENV-3 not protected; DENV-4 too sparse/uncertain.

Efficacy by baseline serostatus

Seropositive and seronegative groups both benefit overall

Outcome	Seropositive VE	Seronegative VE	Appraisal
VCD	64.2% (58.4-69.2)	53.5% (41.6-62.9)	Moderate-high confidence for overall effect
Hospitalised VCD	85.9% (78.7-90.7)	79.3% (63.5-88.2)	Clinically strong but event counts lower
DENV-3 VCD	52.3% (36.7-64.0)	-15.5% (-108.2 to 35.9)	Signal of no efficacy in seronegative participants
DENV-4 VCD	70.6% (39.9-85.6)	-105.6% (-628.7 to 42.0)	Too few seronegative cases for reliable inference

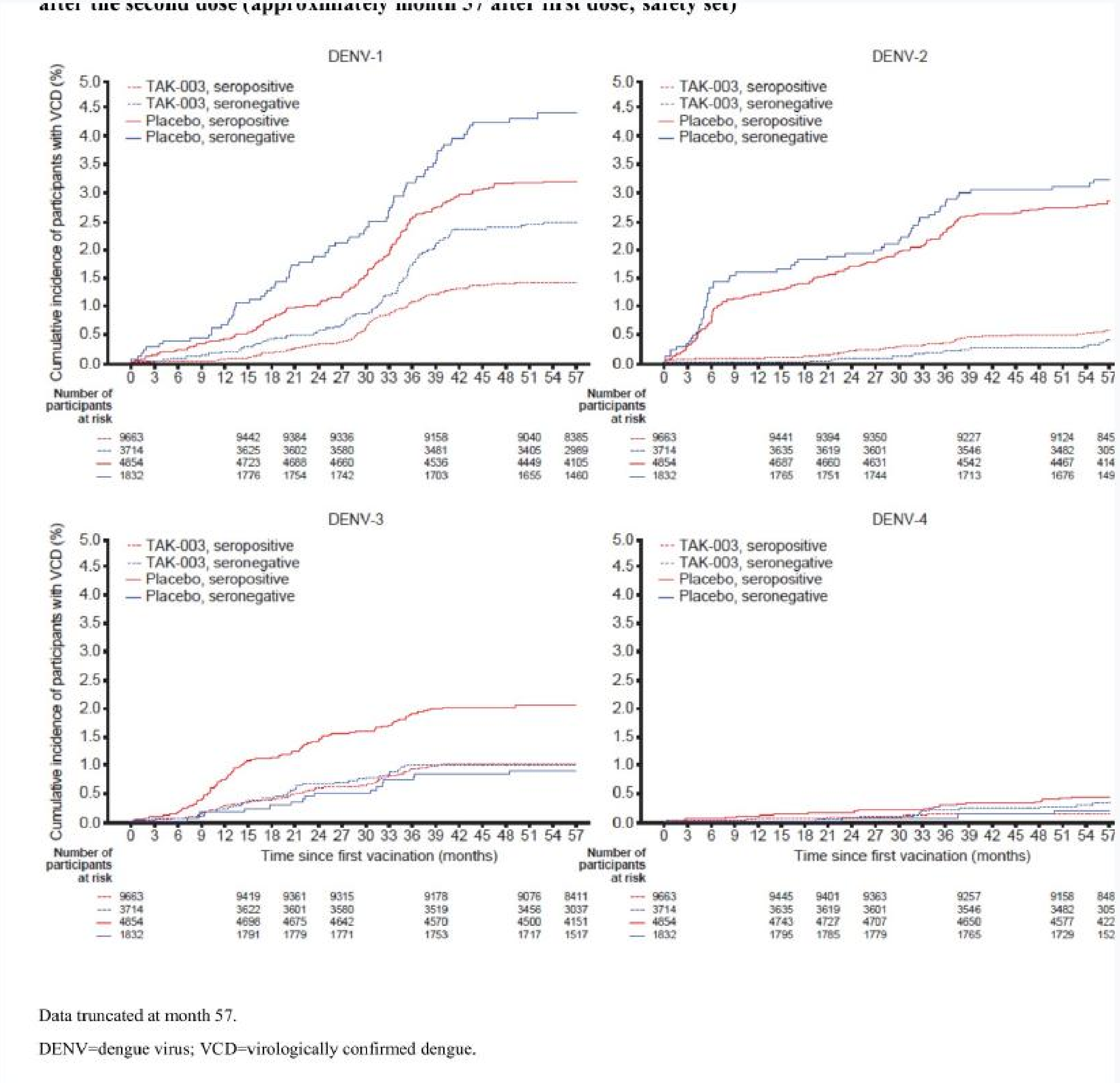
Key clinical distinction

The vaccine looks best established in previously dengue-exposed children/adolescents. It still reduces overall disease in dengue-naive participants, but not evenly across serotypes.

Tricou et al. Lancet Global Health 2024; supplementary appendix 4; CONSORT/RoB 2 appraisal lens.

Serotype-specific signal

The subgroup story drives the benefit-risk debate



Supplement Fig. S2: cumulative VCD incidence by serotype

Pattern from figures and estimates

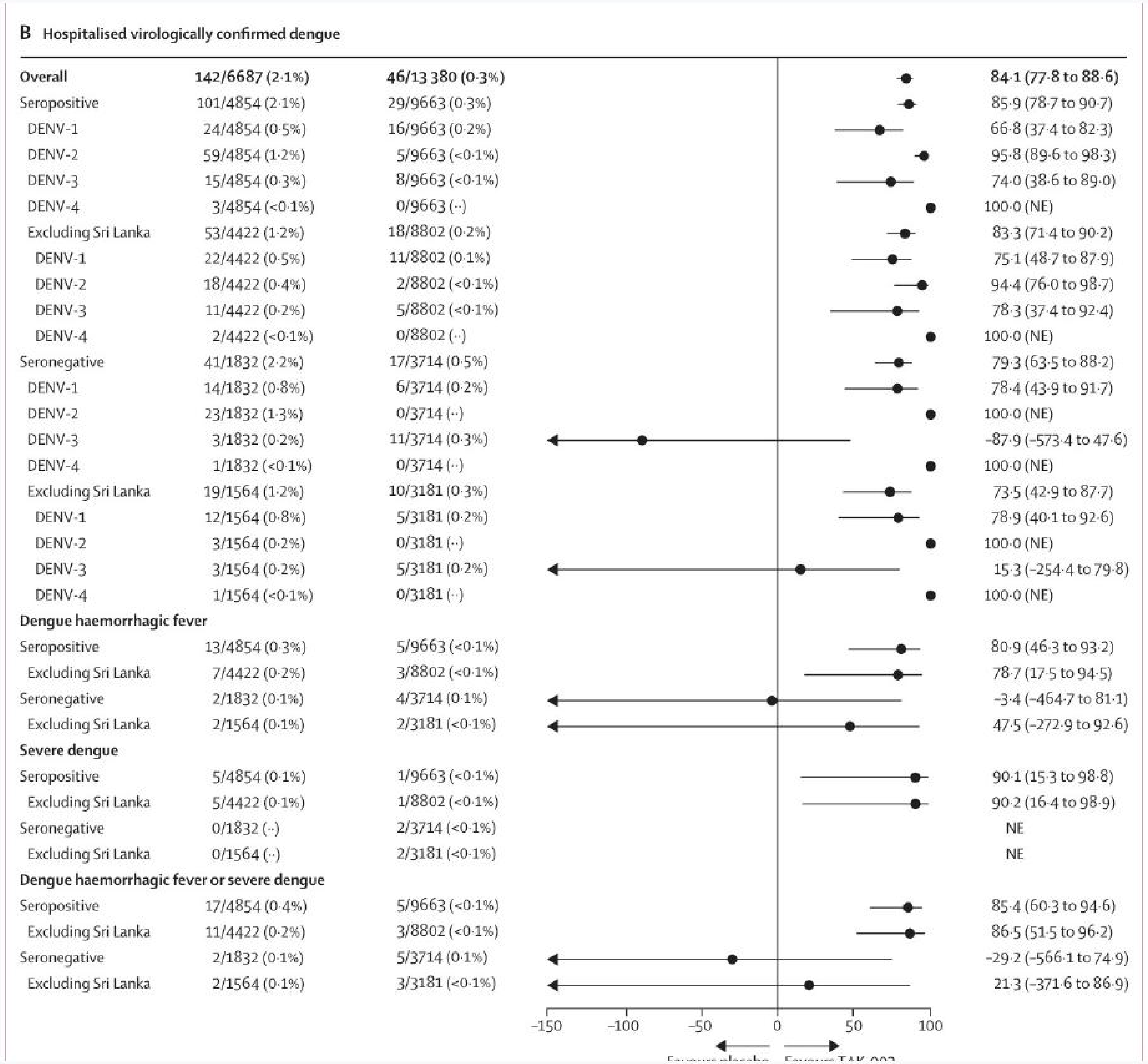
Seropositive: efficacy against all four serotypes.
Seronegative: protection against DENV-1 and DENV-2.
Seronegative DENV-3: no efficacy observed.
Seronegative DENV-4: low incidence prevents meaningful conclusion.

Appraisal

This is not a failure of the overall trial. It is a precision problem created by dengue epidemiology: serotype circulation varies over time and place.

Hospitalised/severe dengue: strong overall, nuanced by setting

Trial figure crop



Interpretation

Overall hospitalised VCD efficacy: 84.1%.
High estimates in both seropositive and seronegative groups.
DENV-3 hospitalised cases in seronegative TAK-003 recipients were numerically imbalanced.
Excluding Sri Lanka reduced some apparent imbalance, but sparse data remain.

Critical caution

Hospitalisation is a patient-important endpoint, but admission practice was heterogeneous across countries.

Durability: benefit persisted into year 4

Not only short-term cross-protection

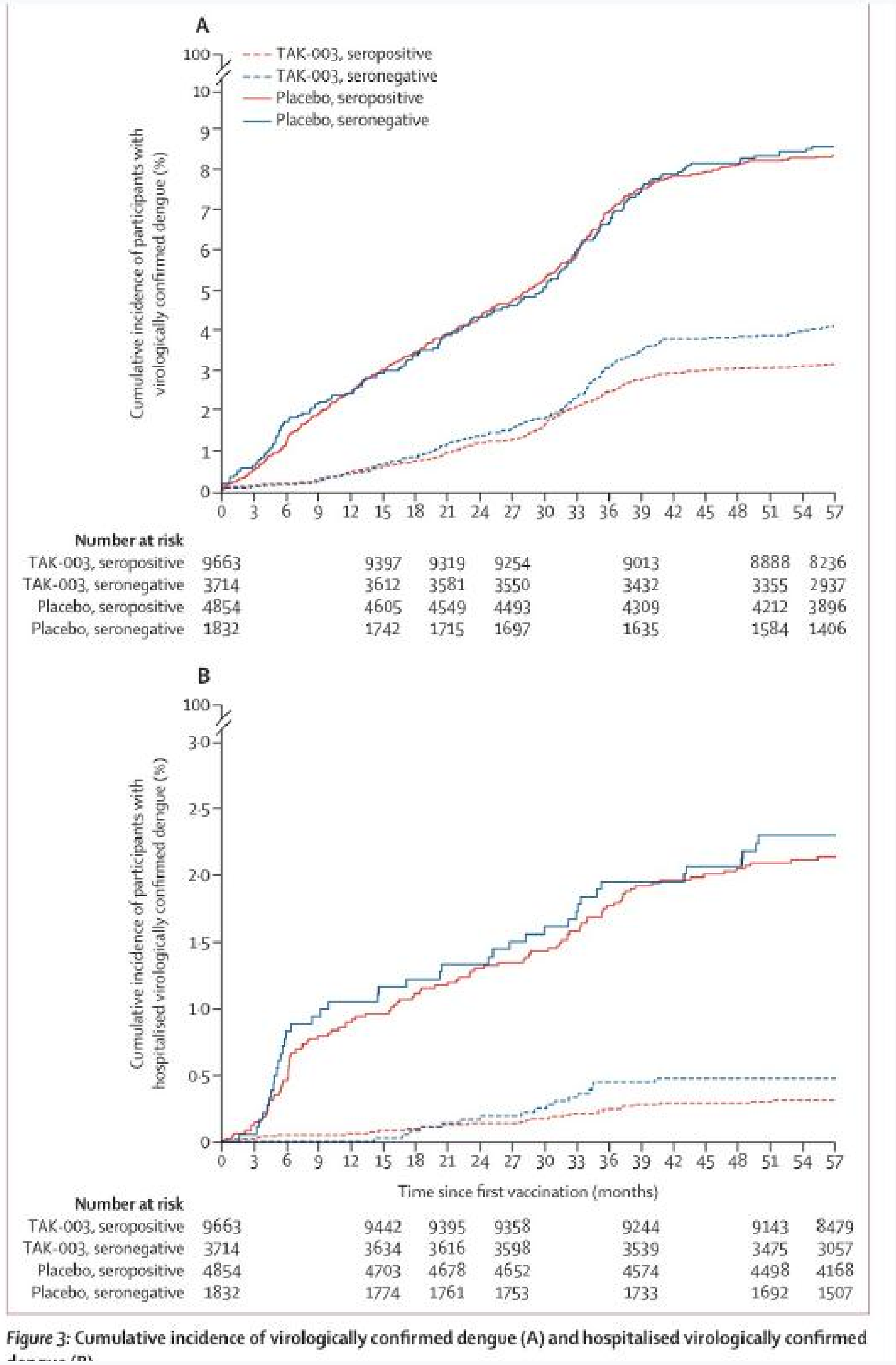


Figure 3: cumulative incidence of VCD and hospitalised VCD

62.8%

VCD efficacy during year 4

96.4%

hospitalised VCD efficacy during year 4

Appraisal

Persistence supports durable protection beyond early immune effects.
But year-by-year estimates fluctuate due to small event counts.
Last 18 months overlapped COVID-19 period and lower dengue incidence.
Booster phase meant not all participants completed identical final follow-up.

Safety findings

No vaccine-related deaths; serious adverse events similar or lower in vaccine arm

Part 3 safety set	Placebo	TAK-003	Comment
Any serious adverse event	396/6687 (5.9%)	664/13380 (5.0%)	No excess SAE signal
Deaths	6	11	In proportion to 2:1 allocation; none vaccine-related
Related serious adverse events	0	0	Investigator assessment
Baseline seronegative any SAE	105/1832 (5.7%)	183/3714 (4.9%)	No overall SAE excess

What safety data support

The long-term trial does not show an overall serious safety signal through 4.5 years.

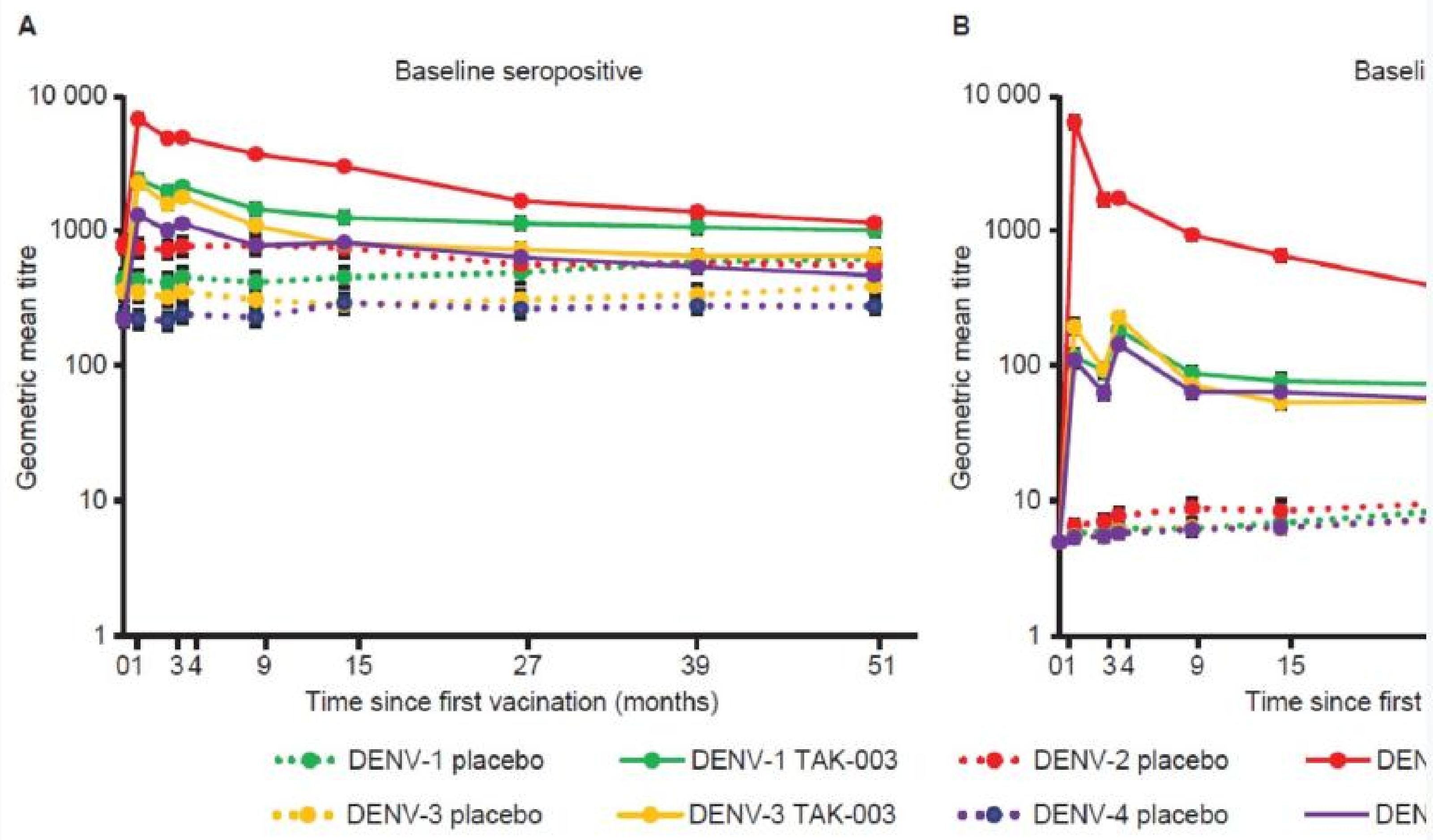
What safety data cannot exclude

Rare vaccine-associated enhanced disease by serotype/serostatus needs larger post-licensure surveillance.

Tricou et al. Lancet Global Health 2024; supplementary appendix 4; CONSORT/RoB 2 appraisal lens.

Immunogenicity: biologic support, not a perfect surrogate

Supplementary figure crop



PPSI analysis set	Placebo	TAK-003
Baseline seropositive	n=902	n=1816
Baseline seronegative	n=345	n=702

ie number of participants evaluated at each timepoint could vary. MNT results were expressed as the reciprocal of the highest dilution of test serum t virus controls (MNT_{50}). DENV=dengue virus. GMT=geometric mean titre. MNT=microneutralisation. PPSI=per protocol set for immunogenicity da

Supplement Fig. S3: serotype-specific GMTs

Appraisal

Neutralising antibody titres remained higher in TAK-003 than placebo through month 51.
DENV-2 titres were particularly high, consistent with DENV-2 backbone.
Immunogenicity supports plausibility but does not fully predict serotype-specific clinical protection.
Clinical efficacy remains the decisive outcome.

Risk of bias: RoB 2 style judgment

Judgment is outcome-specific

Domain	Overall VCD	Hospitalised/severe VCD	Why
Randomisation process	Low	Low	2:1 randomisation, stratified; baseline balance acceptable
Deviations from intended intervention	Low	Low	Double-blind; intervention simple
Missing outcome data	Low/some concerns	Some concerns	91% completion; booster/COVID period may affect late interval precision
Outcome measurement	Low	Some concerns	RT-PCR objective; hospitalisation thresholds varied by country
Selection of reported result	Some concerns	Some concerns	Exploratory endpoints and multiple subgroups
Overall	Low to some concerns	Some concerns	Main efficacy robust; severity/subgroup claims less certain

RoB 2 structure used as a critical appraisal framework, not a formal systematic review extraction.

Internal validity: main strengths

Why the overall estimate is credible

Randomised	Blinded	Objective	Long
allocation with concealment	participants, parents, investigators	RT-PCR confirmation	4.5-year follow-up

Strengths

Large sample size and multicountry recruitment.
Serostatus assessed at baseline, enabling key safety subgroup analysis.
Weekly febrile illness surveillance improves event capture.
Independent dengue severity adjudication committee for severe cases.
Safety monitored throughout the trial.

Why it matters

These design choices reduce selection, performance and detection bias.
They strengthen confidence in the broad efficacy signal.
They do not eliminate sparse-event uncertainty in subgroups.

Tricou et al. Lancet Global Health 2024; supplementary appendix 4; CONSORT/RoB 2 appraisal lens.

Internal validity: main concerns

Where interpretation must slow down

Exploratory analyses
Cumulative 4.5-year analyses and many serotype/serostatus/severity cuts were exploratory.

Multiplicity
Multiple comparisons increase chance findings; type 1 error control was limited beyond key endpoints.

Sparse events
DENV-4 and severe dengue outcomes had too few events for reliable subgroup estimates.

Hospitalisation practice
Sri Lanka had a much higher hospitalization rate for VCD than other countries, affecting clinical severity inference.

Pandemic period
The last 18 months overlapped lower reported dengue incidence during COVID-19 disruptions.

Industry sponsorship
Takeda designed, analysed and funded the study; investigators enrolled and managed participants.

These concerns weaken fine-grained claims more than the main overall efficacy claim

Tricou et al. Lancet Global Health 2024; supplementary appendix 4; CONSORT/RoB 2 appraisal lens.

External validity and implementation

Where can we apply these findings?

Question	Supported by trial?	Comment
Children/adolescents in endemic high-burden areas	Yes	Most applicable population
Age <6 years	Weak	Trial included 4-5, but WHO does not currently recommend programmatic use <6
Adults and elderly	Not directly	Not enrolled in this efficacy trial
Pregnant or immunocompromised persons	No	Excluded; live vaccine cautions apply
Low/moderate transmission settings	Uncertain	Seronegative DENV-3/DENV-4 uncertainty matters more

Policy lens

Current WHO guidance recommends TAK-003 for ages 6-16 in settings with high dengue transmission intensity, and not for programmatic use in low/moderate settings until DENV-3/DENV-4 seronegative uncertainty is better resolved.

WHO Q&A/position paper used for current policy context.

Equity, ethics and public health relevance

Critical appraisal beyond p-values

Ethics

Clinical equipoise existed when the trial was designed.
Children are a vulnerable population; consent/assent and ethics approvals were reported.
Independent monitoring/adjudication supports safety oversight.

Equity/generalizability

Multicountry sites improve diversity of dengue ecology.
Trial participants were healthy; comorbid populations less represented.
Socioeconomic determinants and implementation costs were not central outcomes.

Public health trade-off

A vaccine that reduces hospitalisation can relieve healthcare systems. But use in low-transmission areas may expose more seronegative recipients to uncertain DENV-3/DENV-4 protection.

Tricou et al. Lancet Global Health 2024; supplementary appendix 4; CONSORT/RoB 2 appraisal lens.

What the study proves - and what it does not

Plain-language interpretation for clinicians

Reasonable conclusion	Not established by this trial
TAK-003 reduces overall symptomatic RT-PCR-confirmed dengue in 4-16 year olds in endemic settings.	It does not provide uniform protection against every serotype in dengue-naive participants.
TAK-003 substantially reduces hospitalised VCD overall.	It does not completely settle rare severe disease risk by serotype/serostatus.
Safety through 4.5 years looks favourable in this trial population.	It does not prove safety in pregnancy, immunosuppression, or all adult age groups.
Benefit likely greatest where dengue burden and prior exposure are high.	It does not justify indiscriminate use in low/moderate transmission settings.

Clinical framing: useful tool, not stand-alone dengue control

Tricou et al. Lancet Global Health 2024; supplementary appendix 4; CONSORT/RoB 2 appraisal lens.

GRADE-style certainty summary

Approximate certainty for major claims

Claim	Certainty	Rationale
Overall VCD reduction	High/moderate-high	Large randomized blinded trial; objective RT-PCR outcome; consistent benefit
Overall hospitalised VCD reduction	Moderate-high	Large effect; clinical endpoint; but local admission practice heterogeneity
Seropositive all-serotype protection	Moderate	Positive serotype estimates; subgroup-level inference
Seronegative DENV-1/DENV-2 protection	Moderate	Positive estimates; lower precision than overall
Seronegative DENV-3/DENV-4 protection	Low/very low	No DENV-3 efficacy; DENV-4 sparse; wide intervals
Rare severe safety by serotype/serostatus	Low	Too few severe events; needs post-licensure monitoring

Tricou et al. Lancet Global Health 2024; supplementary appendix 4; CONSORT/RoB 2 appraisal lens.

What would strengthen the evidence?

Next research and surveillance priorities

Post-licensure effectiveness

Large real-world studies in high-burden programmes.
Separate estimates by age, serostatus proxy, region and circulating serotype.
Hospitalisation and severe dengue definitions harmonized across sites.

Safety monitoring

Active surveillance for vaccine-associated enhanced disease.
Serotype-specific breakthrough severity tracking.
AEFI systems able to detect anaphylaxis and rare severe events.

Evidence gaps

Booster dose value and timing.
Effectiveness in adults and comorbid populations.
Performance in low/moderate transmission settings.
Cost-effectiveness integrated with vector control.

Tricou et al. Lancet Global Health 2024; supplementary appendix 4; CONSORT/RoB 2 appraisal lens.

Final bottom line

Critical appraisal synthesis

Take it seriously
This is a well-conducted, large, blinded, multicountry RCT with objective virological confirmation and long follow-up.

Do not overread it
The clinically hard subgroup is dengue-naive recipients by serotype, especially DENV-3 and DENV-4.

Most defensible use
Children/adolescents in high dengue transmission settings, as part of integrated dengue control.

Clinical message
Expect reduction in symptomatic and hospitalised dengue, not sterilizing immunity or equal all-serotype protection.

Overall: favourable benefit-risk in the studied/endemic context, with important seronegative DENV-3/DENV-4 uncertainty

Tricou et al. Lancet Global Health 2024; supplementary appendix 4; CONSORT/RoB 2 appraisal lens.

Key sources

No outside images used; all figure crops are from the uploaded trial article/supplement

Trial and appendix

Tricou V, Yu D, Reynales H, et al. Long-term efficacy and safety of a tetravalent dengue vaccine (TAK-003): 4.5-year results from a phase 3, randomised, double-blind, placebo-controlled trial. Lancet Global Health. 2024;12:e257-e270.
Supplementary appendix 4 for inclusion/exclusion criteria, febrile surveillance, statistical analysis and supplementary figures.

Critical appraisal and policy context

CONSORT 2025 explanation and elaboration: updated guideline for reporting randomized trials.
Cochrane RoB 2: revised tool for assessing risk of bias in randomized trials.
WHO. Vaccines and immunization: Dengue. Q&A updated April 2025.
WHO. Prequalification of TAK-003/Qdenga, May 2024; WHO position paper on dengue vaccines, May 2024.